

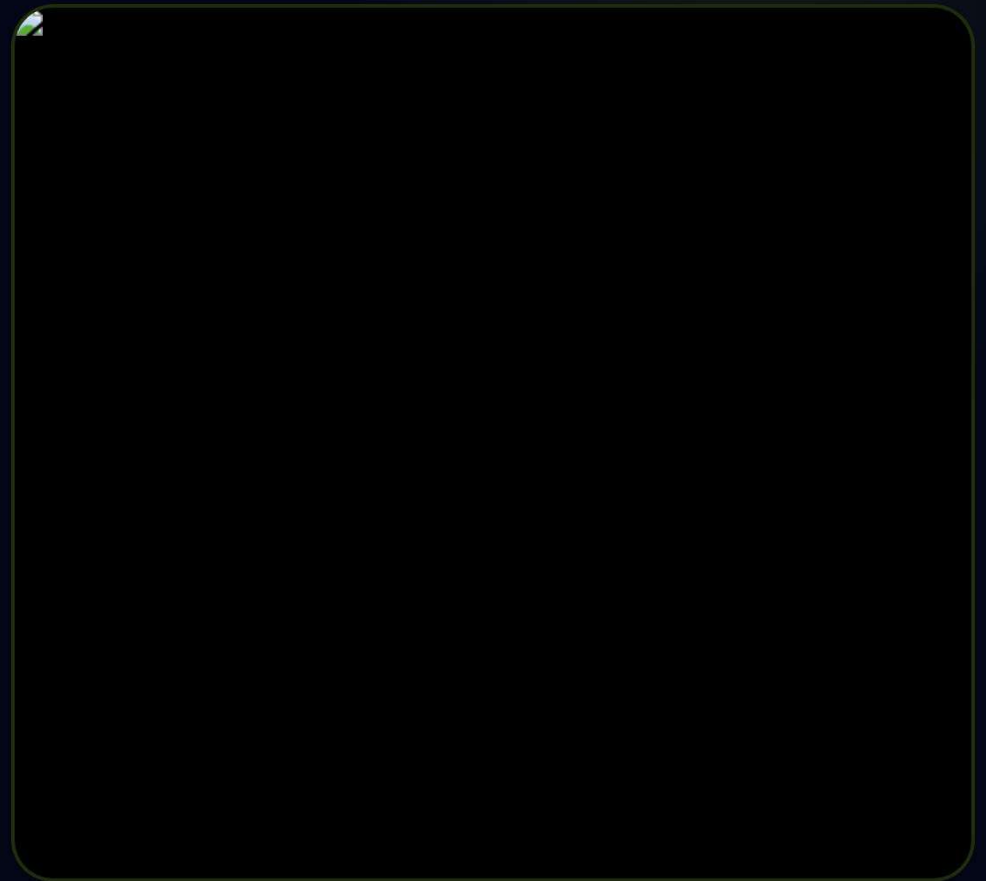
Unitree Go2 AI Evolution

Full-Stack Transformation from Consumer **AIR** to Research-Grade **EDU**
Platform leveraging WebRTC Signaling & Native ROS2 SDK Integration

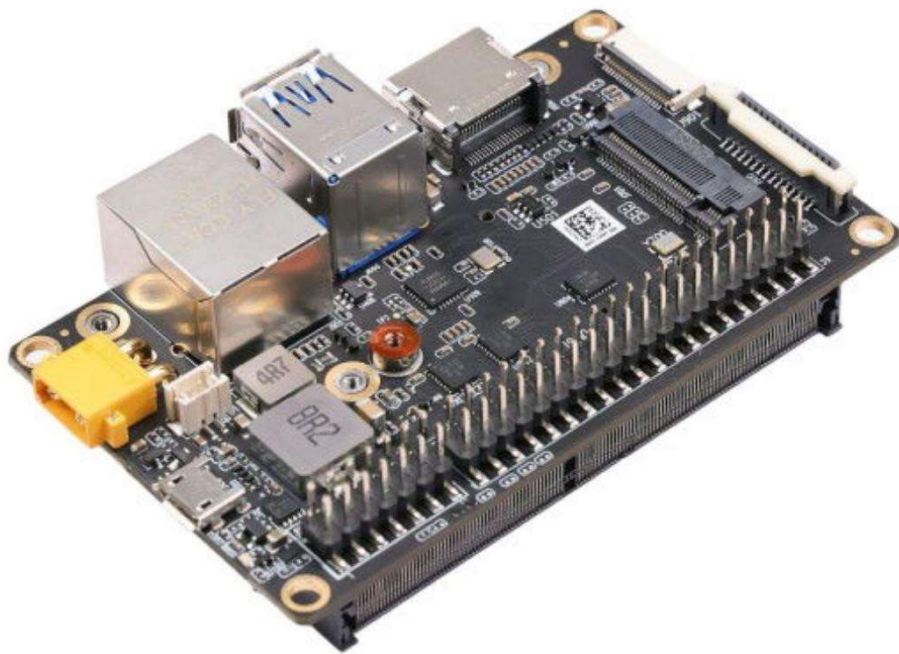
| MISSION: BRIDGING THE GAP

The standard Go2 Air/Pro models are limited by proprietary cloud signaling and app-level interfaces. Our transformation unlocks **Low-Level Control**:

- **Reverse-Engineered Signaling:** Bypassing cloud dependencies via local WebRTC P2P tunnels.
- **SDK Orchestration:** Direct 12-DOF joint-state access and low-latency motor commands.
- **AI Scaling:** Offloading NPU/GPU workloads to the Jetson Orin expansion dock.



HARDWARE: THE INTELLIGENCE STACK



NVIDIA Jetson Orin NX/Nano

Enabling high-performance edge computing on the quadraped's back. This module serves as the primary AI Brain for vision and speech processing.



4D LiDAR-L1 Integration

Utilizing the head-mounted 360° x 90° wide-angle LiDAR for real-time spatial voxel mapping and obstacle awareness.

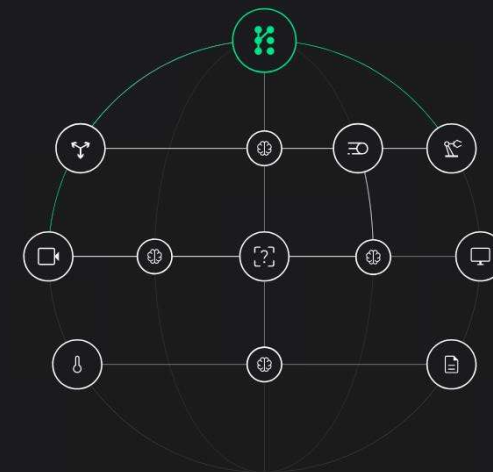
CONTROL: ROS2 SDK CORE

Low-Level RTPS Orchestration

Integrating `go2_ros2_sdk` allows the system to subscribe to low-state topics and publish high-level motion requests directly.

Utilizing **CycloneDDS** middleware to maintain consistent `100Hz` state updates across the robotic cluster.

$$f_{\text{total}} = f_{\text{loop}} + f_{\text{data}} = 200\text{Hz}$$



ROS

WEBRTC CONNECTION STACK

Bypassing standard OTA signaling using reverse-engineered **WebRTC Bridge** logic.

- ✓ **P2P Signaling:** MQTT-based handshake for Local-STA mode.
- ✓ **Data Channels:** Direct tunneling of H.264 video streams.
- ✓ **Remote Access:** Unitree TURN server relay for global control.

Perception Intelligence

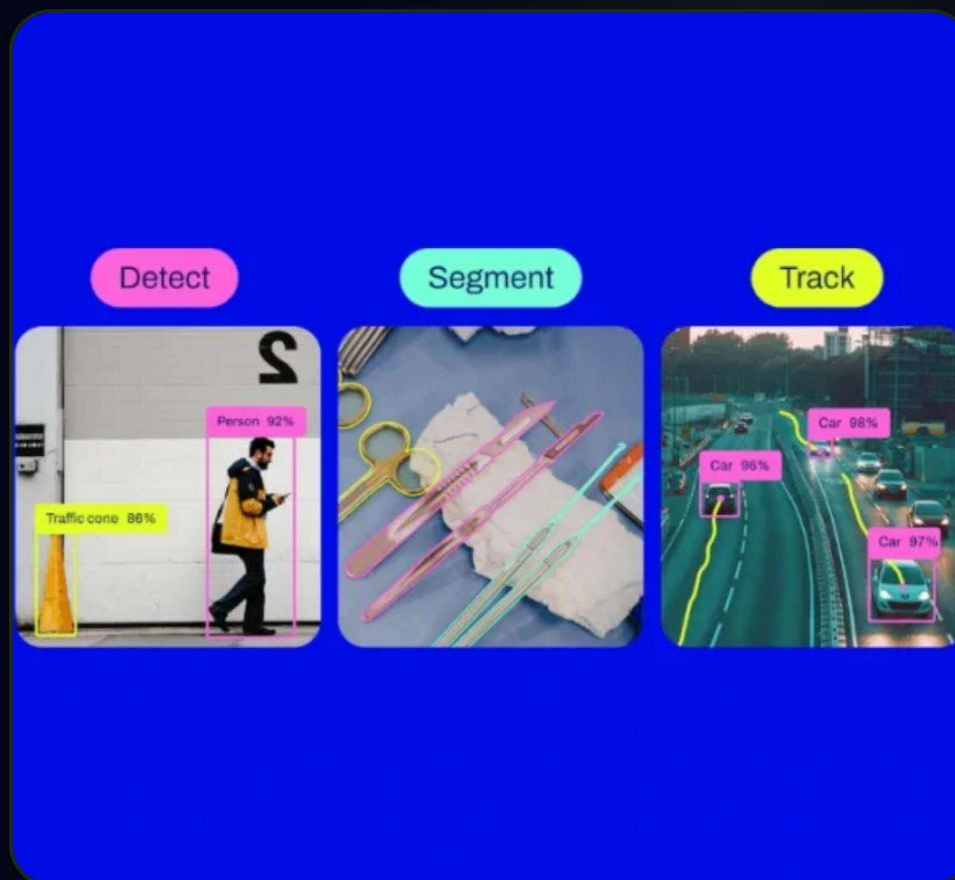
Implementing YOLO-based Vision and 4D LiDAR Spatial Fusion

VISION: YOLO ON TENSORRT

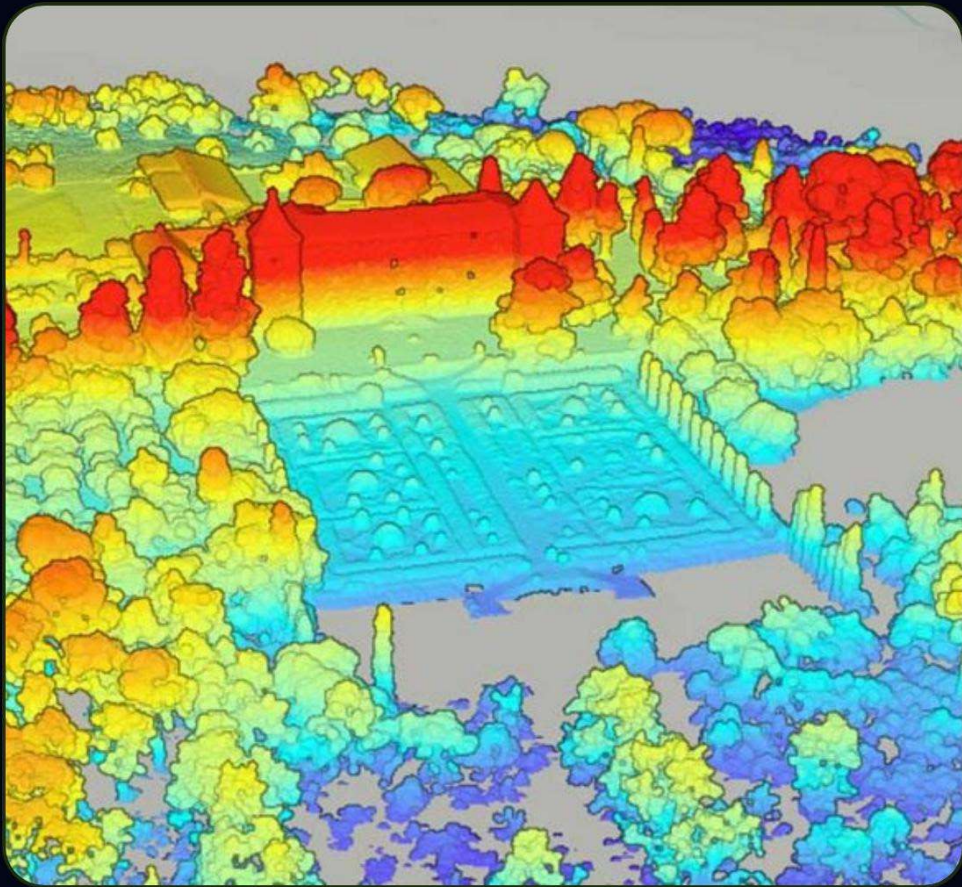
Real-time Neural Inference

Deploying **YOLOv8/v10** for object detection and person re-identification (Re-ID). The model is optimized via **TensorRT** to run at 60+ FPS on the Jetson Orin NX.

- **Zero-Shot Detection:** Using YOLO-World for dynamic target prompts.
- **Centroid Tracking:** Bounding box normalization for motion planning.
- **State Switching:** Adaptive "Search vs. Approach" logic based on confidence.



SPATIAL: 4D LIDAR MAPPING



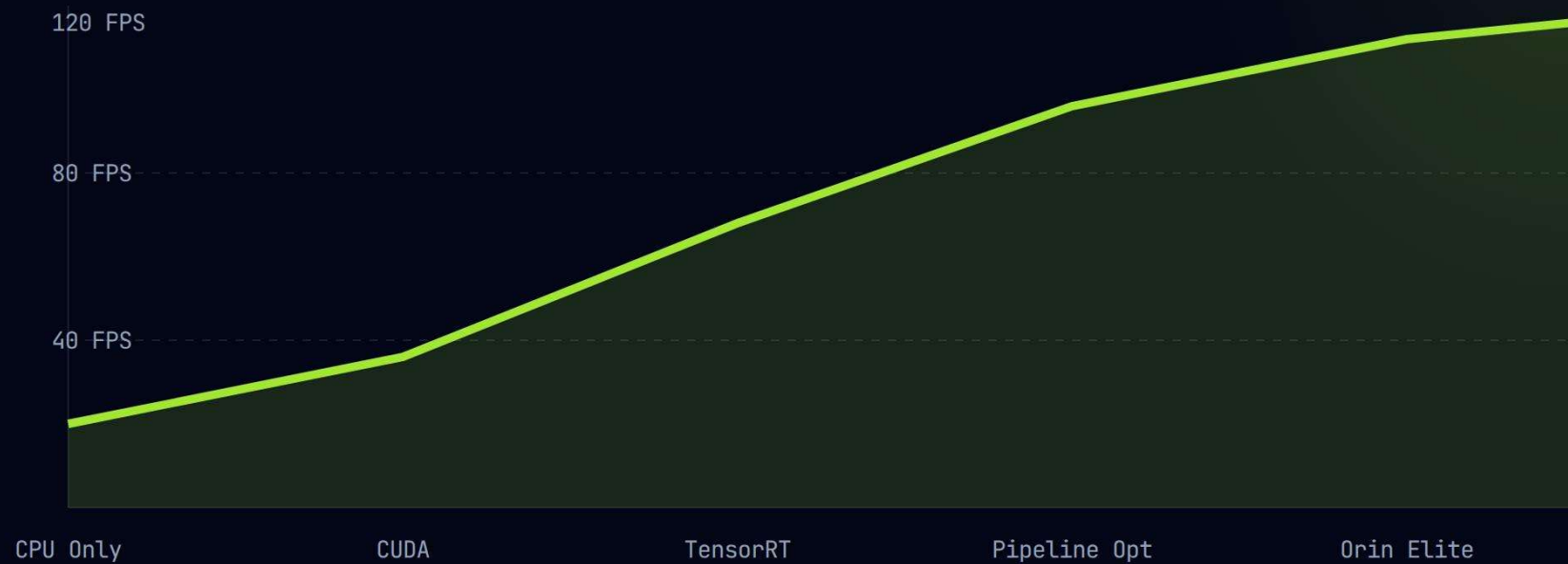
Point-LIO Fusion

Integrating LiDAR-Inertial Odometry for high-frequency state estimation. This provides the robot with "eyes" in 3D space, even in total darkness.

Voxel-Grid Filtering

Real-time obstacle avoidance by projecting YOLO bounding boxes into the 3D voxel map, allowing semantic spatial reasoning.

VISION: PERFORMANCE TREND



System throughput reached 90+ FPS for multi-modal perception pipelines using CUDA/TensorRT acceleration.

Interaction Intelligence

Integrating Google Speech Recognition for Natural Language Control

| INTERACTION: GOOGLE SPEECH API

Human-Robot Conversational Stack

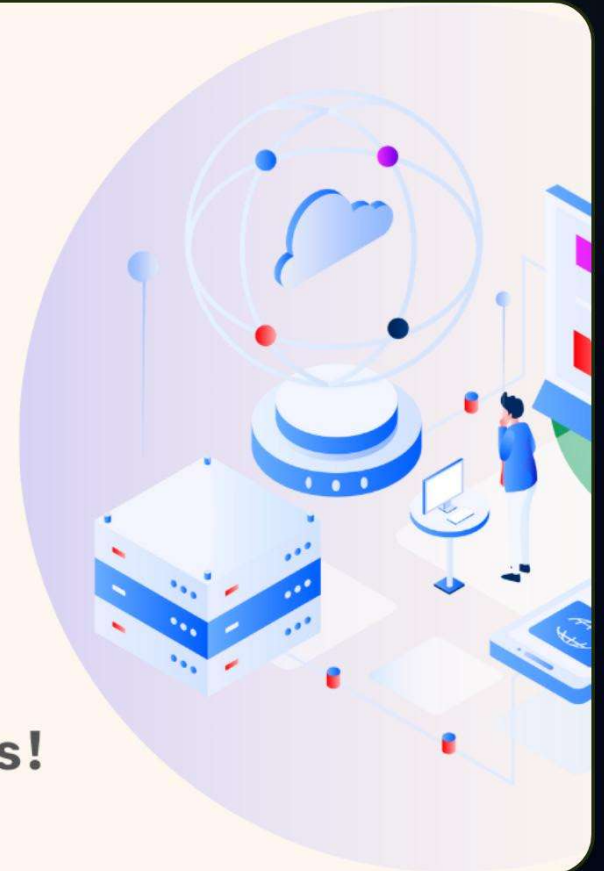
Implementation of **Speech-to-Text (STT)** via Google Cloud Speech-to-Text API. The system captures audio through the Go2 head-mounted mic array.

- **Keyword Activation:** Triggered by customizable hotwords (e.g., "Go2" or "Chloe").
- **Multilingual Support:** Native `en-US` and `zh-CN` recognition.
- **Online Filtering:** Real-time intent validation before command dispatch.

Cloud

's
n-to-Text

estone in AI
gy Applications!



| LOGIC: NLP COMMAND MAPPING

STT to ROS2 Dispatcher

The `nlp_node.py` processes recognized phrases into actionable ROS2 topics.

Example: *"Go to the refrigerator"* → triggers a Nav2 goal to pre-mapped coordinates.

Natural Language Phrase

ROS2 Command Type

"Sit down / Shake hand"

`/rt/api/sport/request`

"What's in the room?"

`YOLO Scan Buffer`

"Stop movement"

`Motion Switcher Stop`

LATENCY: EDGE VS. CLOUD

Operation Module	Standard Cloud Strategy	Enhanced Edge Strategy (Orin)
Vision (YOLOv8)	~200ms (Frame upload req)	~16ms (Local Inference)
Speech (STT)	~800ms (Full buffering)	~250ms (Streaming API)
Motion Control	~150ms (OTA Latency)	< 10ms (Native DDS)
Spatial Mapping	Offline Analysis	Real-time Voxel Mapping

The elite transformation provides up to **10x faster response time** for critical safety behaviors.

SYSTEM ROADMAP



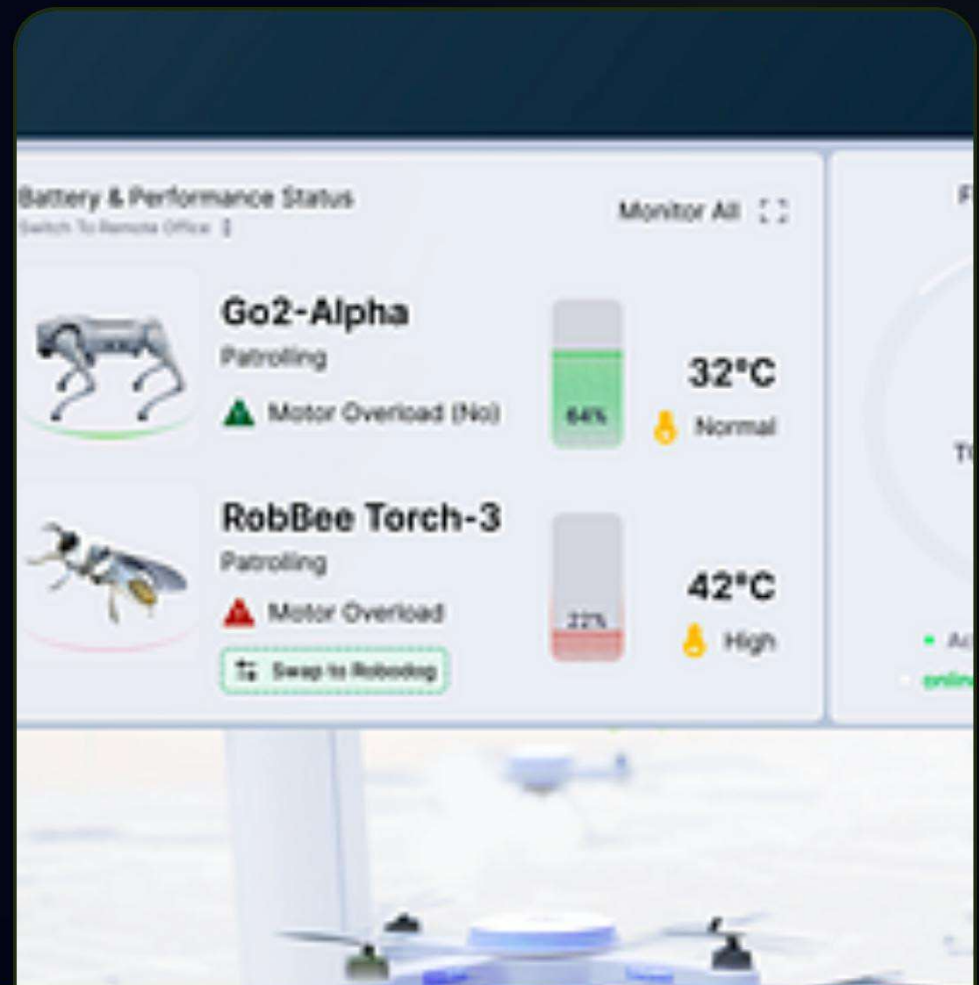
Phase 1: Connectivity

WebRTC reverse engineering and local STA connection established.



Phase 2: Perception

YOLO + Point-LIO fusion for full autonomous mapping.



Transformation Complete.

Unitree Go2 is now an autonomous, vision-aware, voice-responsive robotic research platform.

Questions? Feel free to discuss the implementation details.

Developer: Your Project Lead / Unitree Tech Transformation Team

IMAGE SOURCES

 Thumbnail for <http://robostore.com/cdn/shop/files/unitree-go2-pro-ai-quadruped-robot-dog-4343804.jpg?v=1760102773>
Source: robostore.com

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IMAGE SOURCES



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